

Grammar (unrestricted grammar)

context-free grammar

$$A \rightarrow w$$

grammar

$$uAv \rightarrow w$$

Definition

A grammar (unrestricted grammar) is a 4-tuple

$$G = (V, \Sigma, S, R)$$

- V is an alphabet
- $\Sigma \subseteq V$ is the set of terminals; $V - \Sigma$ " " of non-terminals.
- $S \in V - \Sigma$ is the start symbol
- R , the set of rules, is a finite subset of

$$\underbrace{(V^* (V - \Sigma) V^*)}_{\text{context}} \times V^*$$

1. $(u, v) \in R$, $u \rightarrow v$
2. derives in one step $u \Rightarrow_G v$
3. derives $u \Rightarrow_G^* v$
4. G generates $w \in \Sigma^*$ if $S \Rightarrow_G^* w$
5. $L(G) = \{w \in \Sigma^* : S \Rightarrow_G^* w\}$. G generates $L(G)$.

Observation

A CFG is a grammar.

Example

The following grammar generates $\{a^n b^n c^n : n \geq 1\}$

non-

$$S \rightarrow ABCS$$

$$BA \rightarrow AB, CA \rightarrow AC, CB \rightarrow BC$$

$$S \rightarrow T_c, CT_c \rightarrow T_c c, BT_c \rightarrow BT_b$$

$$BT_b \rightarrow T_b b, AT_b \rightarrow AT_a$$

$$AT_a \rightarrow T_a a, T_a \rightarrow \epsilon.$$

$$S \Rightarrow_G^* \underbrace{ABCABC \dots ABC}_n S$$

$$\Rightarrow_G^* AA \dots AB \dots BC \dots CS$$

$$\Rightarrow_G^* AA \dots AB \dots BC \dots CT_c$$

$$\Rightarrow_G^* A \dots AB \dots BT_c cccc \dots c$$

$$\Rightarrow_G^* A \dots AT_b b \dots bc \dots c.$$

$$\Rightarrow_G^* T_a a \dots a b \dots b c \dots c.$$

$$\Rightarrow_G^* a \dots a b \dots b c \dots c.$$

Theorem

A language is generated by a grammar if and only if it is semidecided by a TM.

Proof:

$\Rightarrow$) Given $G = (V, \Sigma, S, R)$, find a M to semidecide $L(G)$.

$$S \rightarrow ABCS$$

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$BA \rightarrow AB, CA \rightarrow AC, CB \rightarrow BC$

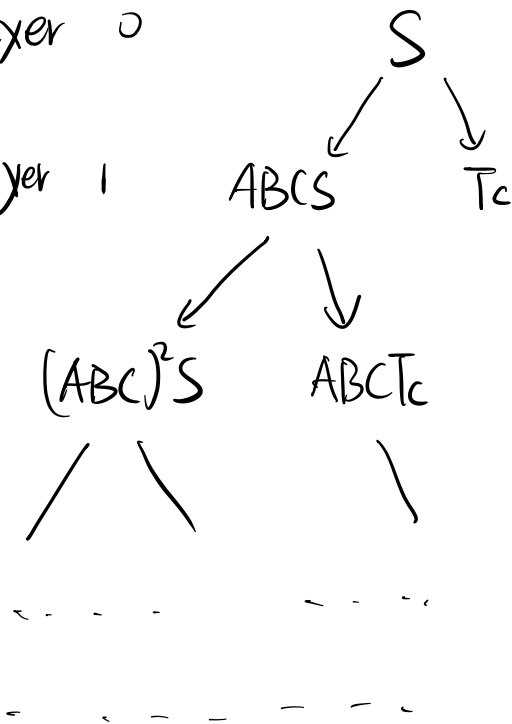
$S \rightarrow T_c, CT_c \rightarrow T_c c, BT_c \rightarrow BT_b$

$BT_b \rightarrow T_b b, AT_b \rightarrow AT_a$

$AT_a \rightarrow T_a a, T_a \rightarrow e.$

layer 0

layer 1



$L(G) = \{ w : w \in \Sigma^* \text{ and } w \text{ is a leaf of the tree} \}$

~~fan~~ fanout $\leq |R|$

nodes in layer $i \leq |R|^i$

M : on input $w \in \Sigma^*$

1. layer 0 = $\{S\}$

2. for $i = 1, 2, 3, \dots$

3. construct layer i from layer $i-1$

4. if w in layer i ,

5. halt

M semidecides $L(G)$.

$\Leftarrow$) Given $M = (K, \Sigma, \delta, s, H)$, find a G to generate $L(M)$.

WLOG, assume that

$$1. K \cap \Sigma = \emptyset$$

2. $H = \{h\}$, and if M halts, it is always in configuration $(h, \triangleright \sqcup)$

$w \in L(M)$

M halts on w

$$(S, \triangleright \sqcup w) \vdash_M (q_1, \triangleright \sqcup_1 q_1 v_1) \vdash_M \dots \vdash_M (q_t, \triangleright \sqcup_t q_t v_t) \vdash_M (h, \triangleright \sqcup)$$

$$\triangleright \sqcup s w \triangleleft \vdash_M \triangleright \sqcup_1 q_1 v_1 \triangleleft \vdash_M \dots \vdash_M \triangleright \sqcup_t q_t v_t \triangleleft \vdash_M \triangleright \sqcup h \triangleleft$$



G generates w

$$\underbrace{S \Rightarrow_G \triangleright \sqcup h \triangleleft}_{\checkmark} \Rightarrow_G \underbrace{\triangleright \sqcup_t q_t v_t \triangleleft}_{\substack{\uparrow \uparrow \uparrow \uparrow}} \Rightarrow_G \dots \Rightarrow_G \triangleright \sqcup_1 q_1 v_1 \triangleleft$$

$$\begin{aligned} &\Rightarrow_G \underline{\triangleright \sqcup s w \triangleleft} \\ &\Rightarrow_G w \end{aligned}$$

$$G = (V, \Sigma', S, R)$$

$$V = K \cup \Sigma \cup \{S, \triangleleft\}$$

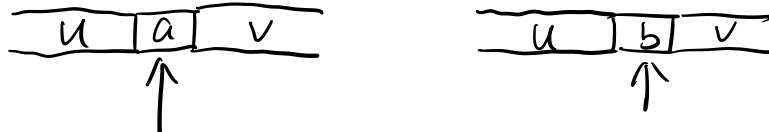
$$\Sigma' = \Sigma - \{\triangleright, \sqcup\}$$

$$R =$$

$$1. S \rightarrow \triangleright \sqcup h \triangleleft$$

$$2. \triangleright \sqcup s \rightarrow e, \triangleleft \rightarrow e$$

$$3. \text{ If } M \text{ has } \delta(q, a) = (q, b) \text{ for } a \in \Sigma, b \in \Sigma.$$

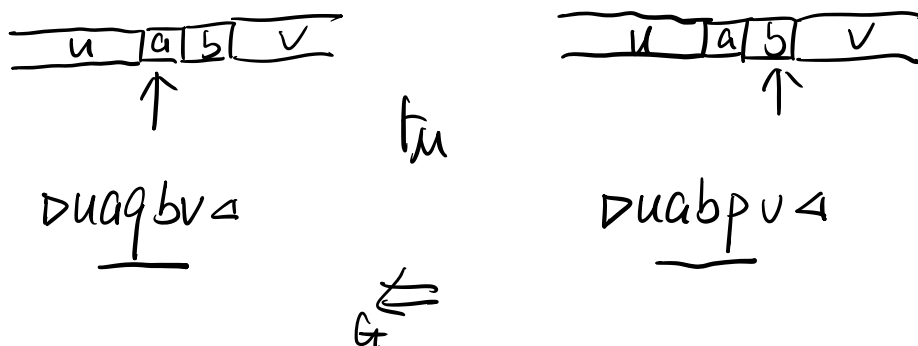


$$\triangleright \underline{u a q v} \triangleleft \quad \vdash_M \quad \triangleright \underline{u b p v} \triangleleft$$

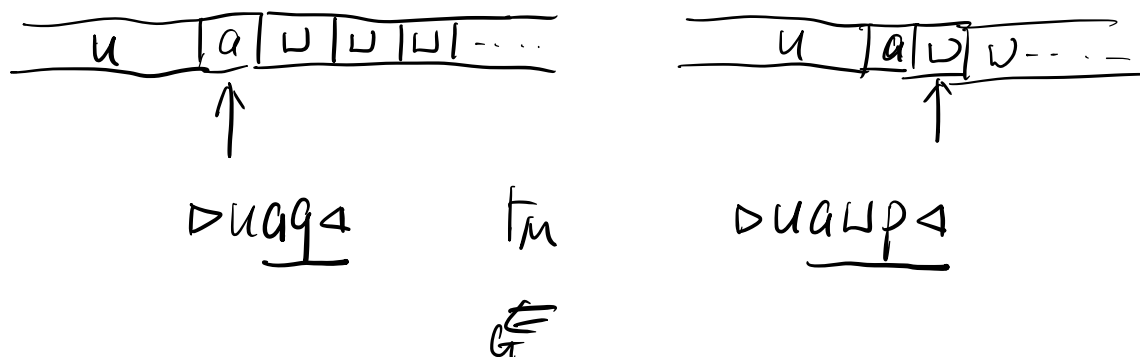
$\stackrel{G}{\Leftarrow}$

G has a rule $bp \rightarrow aq$

4. If M has $\delta(q, a) = (p, \rightarrow)$

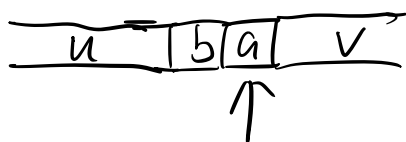


G has a rule $\textcircled{a b p} \rightarrow \textcircled{a q b}$ for each $b \in \Sigma$.
 $\textcircled{b p} \rightarrow \textcircled{q b} \times$



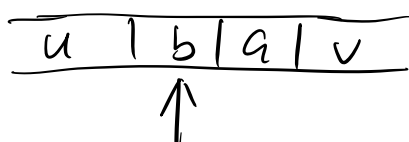
$\textcircled{a \sqcup p} \rightarrow \textcircled{a q}$

5. If M has $\delta(q, a) = (p, \leftarrow)$



DubaqvΔ

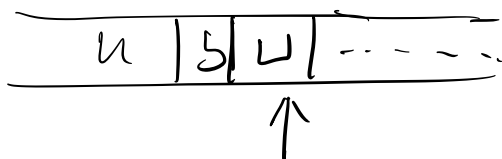
t_M



DubpavΔ

$\xRightarrow{G}$

G has a rule $pa \rightarrow aq$



DubwqΔ

t_M

DubpΔ

$\xRightarrow{G}$

pΔ $\rightarrow$ wqΔ

G generates w iff M halts on w
 $L(G) = L(M)$

Undecidable problems about grammars.

$G \mapsto A_G = \{ \langle G \rangle w : G \text{ is a grammar that generates } w \}$

$H \leq A_G$

$f(\langle M \rangle w) = \langle G \rangle w$

$\langle G \rangle w \in A_G$ iff $\langle M \rangle w \in H$

Rice's Theorem

Let $\mathcal{L}$ be a proper, non-empty subset of the class of all R.E. languages. The following problem is undecidable.

Given a grammar G , is $L(G) \in \mathcal{L}$?

∥

Given a TM M , is $L(M) \in \mathcal{L}$? undecidable

Theorem

$ND_{CFG} = \{ \langle G_1, \langle G_2 \rangle : G_1 \text{ and } G_2 \text{ are two CFGs with } L(G_1) \cap L(G_2) \neq \emptyset \}$

Proof:

$A_G \leq ND_{CFG}$

$f(\langle G \rangle \langle w \rangle) = \langle G_1, \langle G_2 \rangle \rangle$ s.t. $\langle G \rangle \langle w \rangle \in A_G$ iff $\langle G_1, \langle G_2 \rangle \rangle \in ND_{CFG}$.

reduction by computation history.

$\langle G \rangle \langle w \rangle \in A_G$ where $G = (V, \Sigma, S, R)$

$\Leftrightarrow G$ generates w

$\Leftrightarrow \exists$ a derivation $S \Rightarrow_G u_1 \Rightarrow_G u_2 \Rightarrow_G \dots \Rightarrow_G w$ of length $n \geq 1$

$\Leftrightarrow \exists$ a string, $u_0 \# u_1 \# u_2 \# \dots \# u_{n-1} \# u_n$,

s.t. $u_0 = S$, $u_n = w$, $n \geq 1$, and $u_i \Rightarrow_G u_{i+1}$ for $i=0, \dots, n-1$

$\Leftrightarrow \exists$ a string, $u_0 \# u_1^R \# u_1 \# u_2^R \# u_2 \# \dots \# u_{n-1}^R \# u_{n-1} \# u_n^R$

s.t. $u_0 = S$, $u_n = w$, $n \geq 1$, and $u_i \Rightarrow_G u_{i+1}$ for $i=0, \dots, n-1$

$\Leftrightarrow \exists$ a string $u_0 \# v_1 \# u_1 \# u_2 \# u_2 \# \dots \# v_{n-1} \# u_{n-1} \# v_n$

s.t. $n \geq 1$, $u_0 = S$, $v_n = w^R$, $v_i = u_i^R$, $u_i \Rightarrow_G v_{i+1}^R$

$\Leftrightarrow L(G) \cap L(G) \neq \emptyset$

$\Leftrightarrow L(G_1) \cap L(G_2) \neq \emptyset$
 CFG
 $\Leftrightarrow "G_1", "G_2" \in \text{NPCFG}$

$$L(G_1) = \{ \underline{u_0 \# v_1 \# u_1 \# v_2 \# u_2 \#} \dots \# v_{n-1} \# u_{n-1} \# v_n \mid u_i \Rightarrow_G v_{i+1}^R, n \geq 1 \}$$

$$\begin{aligned}
 S_1 &\rightarrow P \# S_1 \\
 P &\rightarrow aPa \text{ for } a \in \Sigma \\
 P &\rightarrow \alpha P' \beta^R \text{ for } (\alpha, \beta) \in R \\
 P' &\rightarrow aP'a \text{ for } a \in \Sigma \\
 P' &\Rightarrow \#
 \end{aligned}$$

$$\begin{aligned}
 u_i &\Rightarrow_G v_{i+1}^R \\
 u\alpha v &\Rightarrow_G u\beta v \quad (\alpha, \beta) \in R \\
 u_i \# v_{i+1} \\
 \underline{u\alpha v \# v\beta^R u}
 \end{aligned}$$

$$\begin{aligned}
 \hookrightarrow L(G_2) &= \{ u_0 \# \underline{v_1 \# u_1 \# v_2 \# u_2 \#} \dots \# v_{n-1} \# u_{n-1} \# v_n \mid \\
 &\quad v_i = u_i^R, u_0 = S, v_n = w^R, n \geq 1 \}
 \end{aligned}$$

$$\begin{aligned}
 S_2 &\rightarrow S \# S_2' \# w^R \\
 S_2' &\rightarrow P \# S_2' \quad abc \# cba \\
 P &\rightarrow aPa \text{ for } a \in \Sigma \\
 P &\rightarrow \#
 \end{aligned}$$

$$\text{All}_{\text{CFG}} = \{ "G" \text{ is a CFG with } L(G) = \Sigma^* \}$$

$$\text{EQ}_{\text{CFG}} = \{ "G_1", "G_2" \text{ are two CFGs with } L(G_1) = L(G_2) \}$$